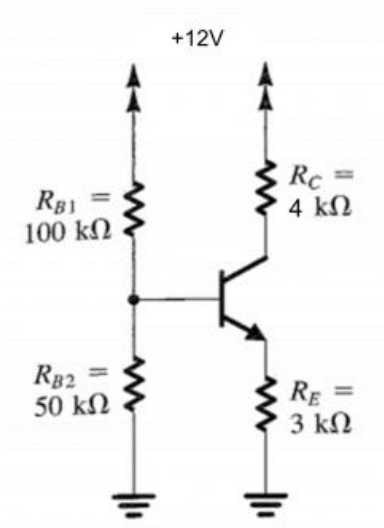
BJT Assignment Problems:

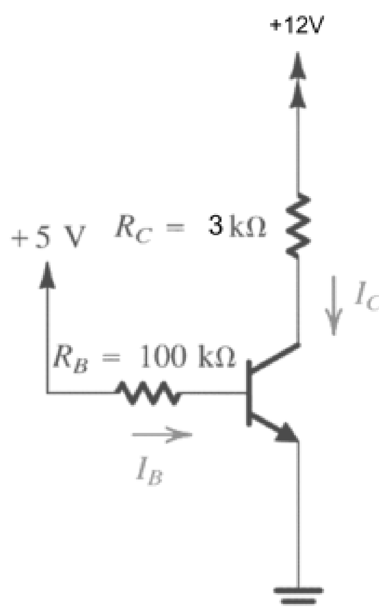
1.

For the transistor in the following circuit, $\beta = 50$ and $\beta_{forced} = 15$. Analyze the circuit to determine the voltage values at all nodes and the current through all branches.



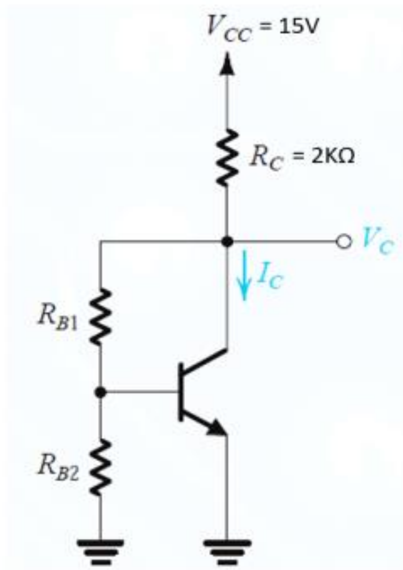
2.

Find all node voltages and branch currents in the following circuit, $\beta = 155$



3.

Q3. For the circuit shown in Figure 4, the transistor parameters are RB1 = RB2 = 50KΩ, $V_{BE}(on) = 0.7$ V and $\beta = 100$. Find I_B, I_C, I_E and V_{CE}. **[7]**



4.

4. Find all node voltages and branch currents in the following circuit in Fig. 5. Assume, β=100 and $R_{B1} = R_{B2} =$ 250KΩ. **[7]**

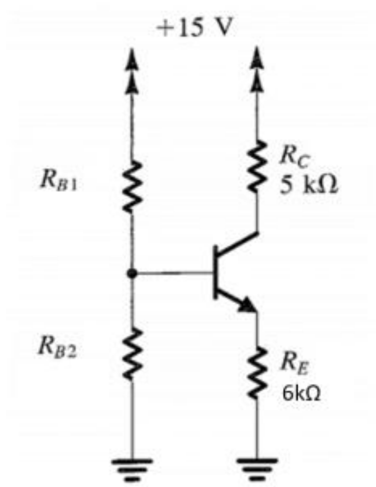
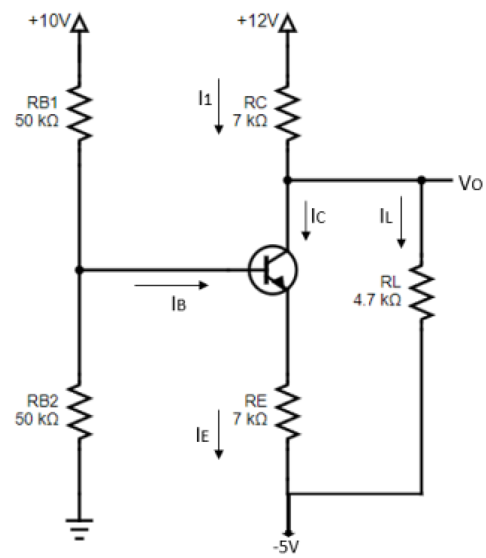


Fig 5: Circuit diagram for Q4

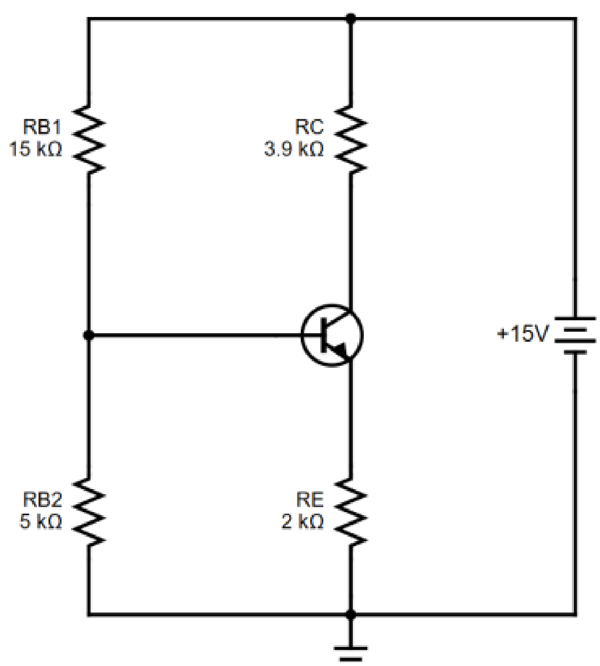
5.

Q3. For the circuit shown in Figure 4, the transistor parameters are: $V_{BE}(on) = 0.7$ V and $\beta = 100$. Find I_B, I_C, I_L, I_1, I_E, V_O and V_{CE}. **[8]**



6.

Q4. Find all node voltages and branch currents in the following circuit in Fig. 5. Assume, β=100 and $\beta_{forced} = 25$. **[7]**



7.

Q3. Determine the voltages at all nodes and current through all branches of circuit in Fig. 4. Given, the current gain, $\beta = 110$. **[5]**

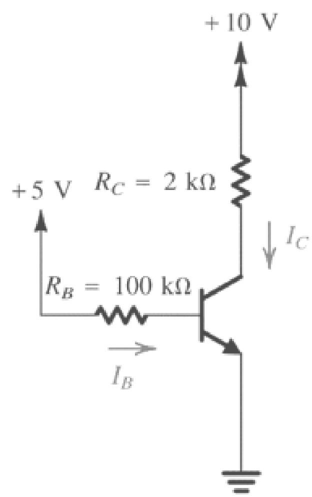


Fig 4: Circuit diagram for Q3

8.

Q4. Find all node voltages and branch currents in the following circuit in Fig. 5. Assume, $R_{B1} = R_{B2} = 50\ k\Omega$ and $\beta=100$.

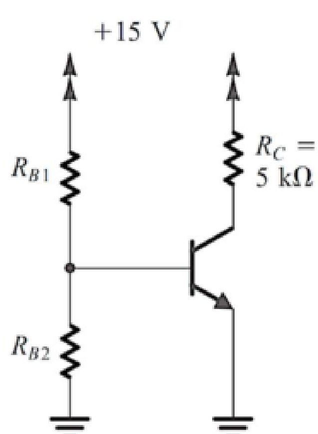
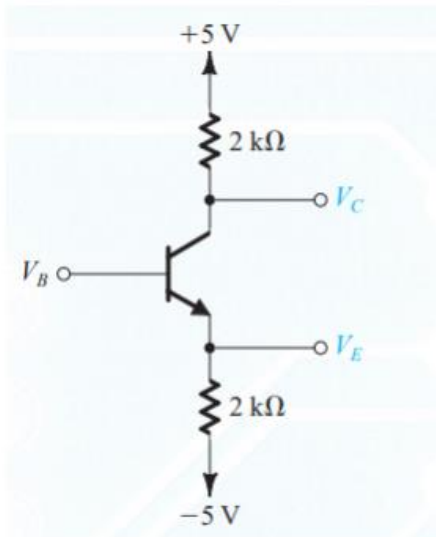


Fig 5: Circuit diagram for Q4

9.

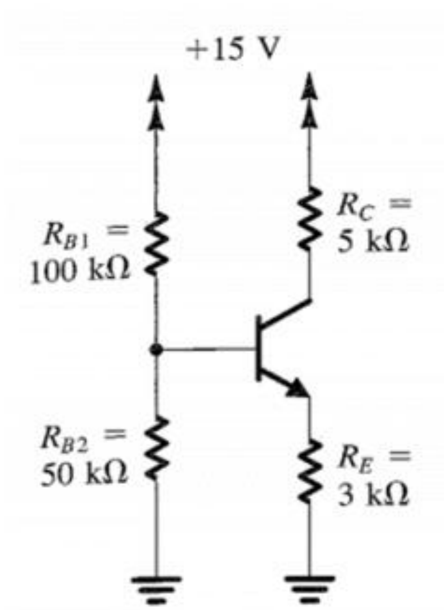
Q3. The transistor shown below is operating at the edge of saturation. Find the values of voltages at all nodes (V_B, V_C, V_E) and the currents through all branches.

Here, $\beta = 100$. [4]



10.

Q4. For the transistor in the following circuit, $\beta = 50$ and $\beta_{forced} = 15$. Analyze the circuit to the value of voltage at all nodes and current through all branches.[5]



11.

Q3. Determine the voltage at all nodes and current through all branches for the BJT circuit shown below. Here, $R_E + R_C = 8\ k\Omega$ and $0 < R_E < R_C$ [4]

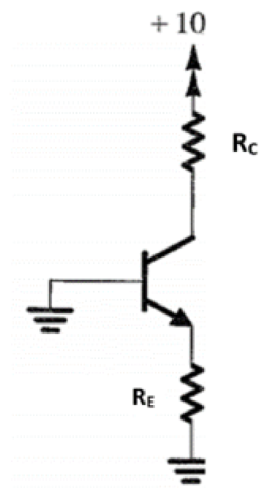
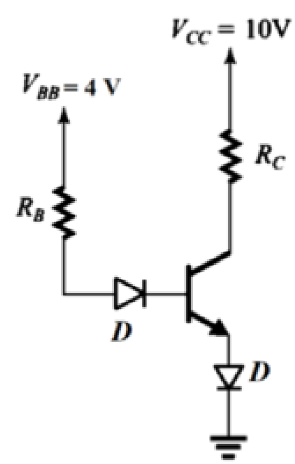


Fig 4: Circuit diagram for Q3

12.

Q4. For the bipolar junction transistor in the following circuit,

R_B=100 kΩ and $\beta_{forced} = 15$. Find the value of R_C. Assume silicon diode. [6]



13.

(a) Given $V_B = 4$ V for the network of the following figure, determine $V_E, V_C, I_C, I_B, I_C, I_E$ and β. [7]

(b) Find the value of V_0 from the following figure. [4]

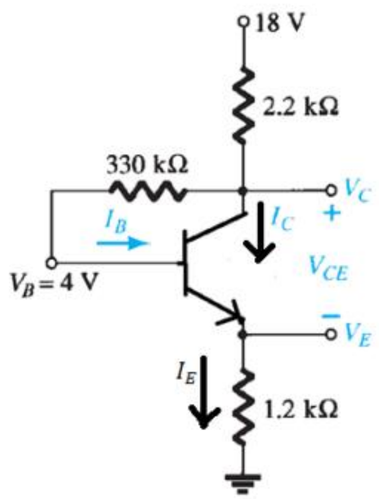


Figure 3: Circuit diagram for Q-2(a)

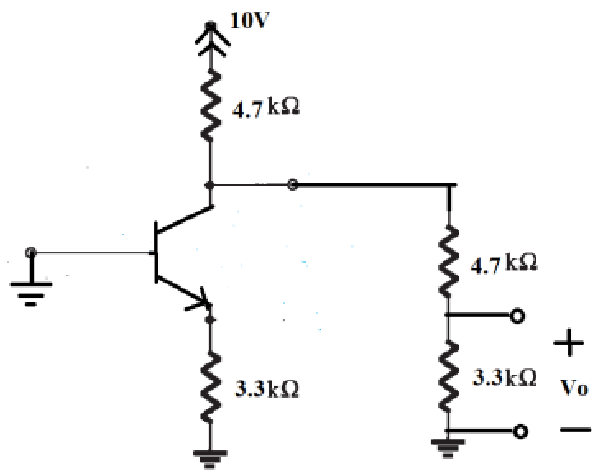


Figure 4: Circuit diagram for Q-2(b)

14.

5. (a) For the circuit in **Figure 4.**, it is required to determine the value of the voltage that results in the transistor operating at the edge of saturation. The transistor β is specified to be 50. **[5]**

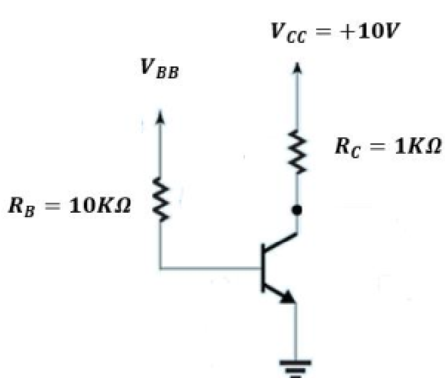


Figure 4.

15.

(b) From the circuit in **Figure 5.**, determine the values of $V_C, V_E, V_B, I_C, I_B, and\ I_E$. Assume that $R_B = 10K\Omega$. **[6]**